

If/Then Decisions

Project: Helpful Classroom Sorter

1. Your mission

Your learning target

I can evaluate a condition using the current state and trace only the selected branch.

I can distinguish inclusive and strict boundaries using values below, at and above a threshold.

I can design a boundary test and handle an unknown input without silently treating it as false.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

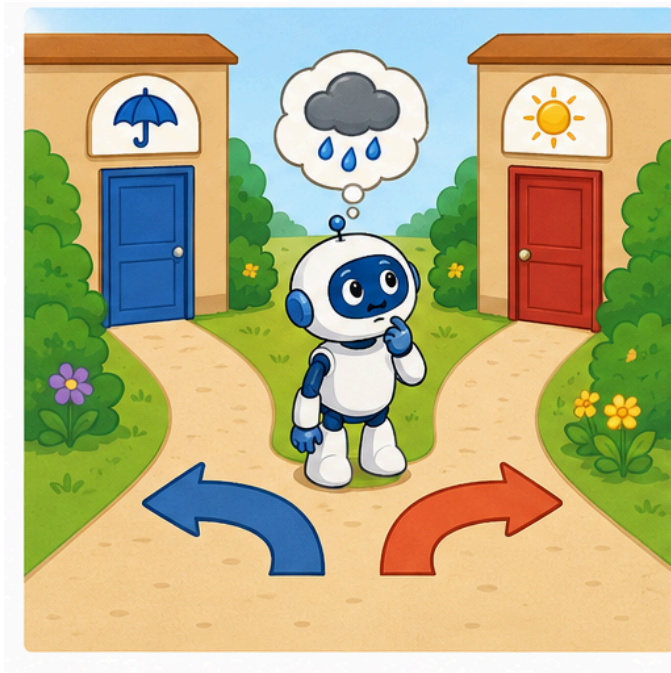
Without opening the old lesson, explain one key idea from Algorithms as Recipes. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Categories and Labels. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Rules or Learned Patterns?. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Conditions make an algorithm branch when information changes.



LEARNING TARGET A club admits a visitor if they have a ticket AND an accompanying adult. Test all four yes/no combinations. A student changes AND to OR. Which outcomes change?

Learner name:

Readiness for the connected explanation

Score starts at 4. Add 2 to score, then double it. Which value is doubled, and what is the final score?

This week's end goal

By the end, I can trace exactly one branch and test an including equality boundary.

2. Notice the evidence

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; evidence tells us what the clues mean.



MYSTERY A clothing helper follows different paths depending on temperature, rain, and wind conditions.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first prediction and state the clue that supports it.

My observations, prediction, and evidence:

GOOD THINKING A prediction can change when stronger evidence appears.

A closer explanation

A paper game updates a score with this procedure: add 2 to score; if score is at least 6, subtract 3 from score; otherwise add 4 to score; report score and stop. The if and otherwise instructions form one if/else decision. Exactly one of those branches runs each time this decision is reached.

A condition asks whether a statement is true for the current state. At least 6 means 6 or any larger number. If score is 6, the condition is true. If score is 5, it is false. The condition itself does not change score; the chosen branch performs the update.

Start score at 4. First add 2, giving 6. Now test the current 6: the condition is true, so subtract 3 and report 3. Testing the original 4 would choose the wrong branch. Write the state immediately before the condition so the evidence for the choice is visible.

3. See how it works

VISUAL MECHANISM

Input → **Process** → **Output**

Follow the information instead of imagining magic inside the machine.



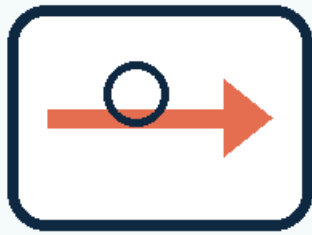
INPUT

weather values



PROCESS

check ordered if/then conditions



OUTPUT

clothing suggestion and reason

INPUT weather values

PROCESS check ordered if/then conditions

OUTPUT clothing suggestion and reason

CORE IDEA Conditions make an algorithm branch when information changes.

UNDER THE HOOD A conditional evaluates a Boolean expression and selects a branch. Order matters when conditions overlap. Complete decision logic needs explicit handling for boundary values and an else path.

Draw a fresh example with all three stages:

A closer explanation

Compare the three values immediately around the threshold. These are the values when the condition is reached, after the initial add-2 instruction. Testing only a value far above 6 would not show whether equality was handled correctly.

Three checks around the threshold

Score at the condition · Choose one branch · Score after the branch

Score at the condition: Score is 5

Choose one branch: At least 6 is false so add 4

Score after the branch: Score becomes 9

Score at the condition: Score is 6

Choose one branch: At least 6 is true so subtract 3

Score after the branch: Score becomes 3

Score at the condition: Score is 7

Choose one branch: At least 6 is true so subtract 3

Score after the branch: Score becomes 4

At least includes equality. The middle row is essential for checking this boundary. Read each row as a separate run; do not pass the result of one row into the next.

Above 6 is different: it means strictly greater than 6, so 6 itself does not pass. At most 6 means 6 or smaller. Below 6 means strictly smaller, so it excludes 6. Say the equality case aloud before using one of these phrases.

A condition can be true even when its chosen action makes the number smaller. True describes the test result; it does not mean good, bigger, or a successful game. The written instructions define the action.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

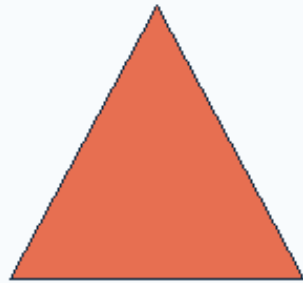
PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the evidence carefully.



Rain = yes



Temperature < 15 C



Exactly 15 C



Rain and cold

CASE 1 — Rain = yes Clue: then choose raincoat. Meaning: clear binary branch.

CASE 2 — Temperature < 15 C Clue: then choose warm layer. Meaning: numeric threshold.

CASE 3 — Exactly 15 C Clue: needs boundary rule. Meaning: edge case.

CASE 4 — Rain and cold Clue: two conditions overlap. Meaning: order or combined rule needed.

BOUNDARY Never hide an unhandled case between two thresholds.

Which two pictures are easiest to confuse? What evidence separates them?

Change one thing in the pictured case

Change: Replace at least 6 with above 6.

Prediction: Only the run starting at 4 changes: it reports 12.

Reason: Updated value 6 fails a strict above comparison and is doubled.

5. Words that unlock the idea

VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.





branch



threshold



Word · Meaning · My example

Word: condition

Meaning: question evaluated as true or false

My example: _____

Word: branch

Meaning: path selected by a condition

My example: _____

Word: threshold

Meaning: value where a decision changes

My example: _____

Word: else

Meaning: path used when earlier conditions are false

My example: _____

Misconception detective

CLAIM TO REPAIR If one example works, the branch is complete.

Use one vocabulary word and one piece of evidence to repair it:

WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

Word · Meaning

Word: Inclusive boundary

Meaning: A boundary whose equality case is included

Word: Boundary test

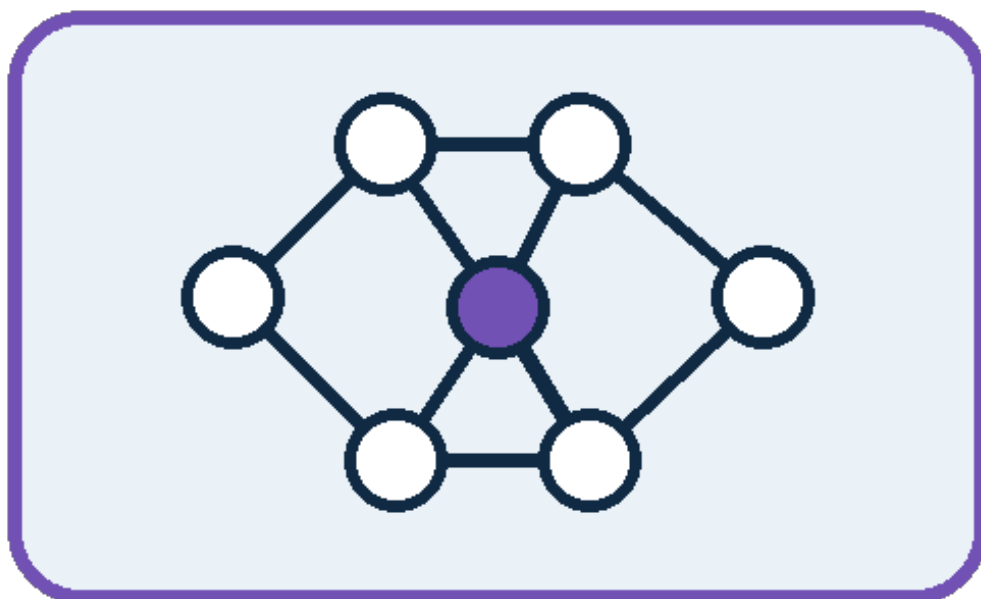
Meaning: A test using a value at or close to a point where behaviour can change

6. Follow a worked example

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable evidence.



CASE A clothing helper follows different paths depending on temperature, rain, and wind conditions.

STEP 1 — OBSERVE weather values

STEP 2 — TRACE check ordered if/then conditions

STEP 3 — CONCLUDE clothing suggestion and reason

MODEL REASONING Each test follows the stated condition order. The exactly-at-threshold case reveals whether the rule uses less than or less-than-or-equal. Rain-plus-cold needs combined outputs or a clear priority.

DO NOT OVERCLAIM Never hide an unhandled case between two thresholds.

What new evidence could change this answer?

A closer explanation

Initial score · State and condition after adding 2 · Branch and final score

Initial score: 3

State and condition after adding 2: 5 is not at least 6

Branch and final score: Add 4 and report 9

Initial score: 4

State and condition after adding 2: 6 is at least 6

Branch and final score: Subtract 3 and report 3

Initial score: 5

State and condition after adding 2: 7 is at least 6

Branch and final score: Subtract 3 and report 4

In the first run, the false branch increases 5 to 9. That does not make the earlier decision run again. The procedure has already chosen and carried out the otherwise branch, so it moves on to report and stop. Subtracting 3 afterward would incorrectly run both branches and report 6.

This rule applies to one if/else decision. A later, separate condition would be a new test of the state at that later point. Do not invent an extra test or loop because the value crossed the threshold. Follow only the

instructions in the procedure.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: If/Then Decisions

A fictional counter begins with a number, adds 2, then chooses one branch.

The facts we will use

If the updated value is at least 6, subtract 1; otherwise double it.

Report once and stop. Use separate starting values 3,4,5.

AI ACADEMY / AI-1/10

Inclusive branching trace

	Start	After +2	At least 6?	Report
Run 1	3	5	false	10
Run 2	4	6	true	5
Run 3	5	7	true	6

Read a cell using both its row and column labels.

If/Then Decisions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Update first

Starts 3,4,5 become 5,6,7.



2. Test current value

$5 \geq 6$ is false; $6 \geq 6$ and $7 \geq 6$ are true.



3. Choose one branch

Double 5 to 10; subtract 1 from 6 to 5; subtract 1 from 7 to 6.



4. Report

The three independent runs report 10,5,6.

CHANGE ONE THING

Replace at least 6 with above 6.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Update first

What happens and why: Starts 3,4,5 become 5,6,7.

Step: 2. Test current value

What happens and why: $5 \geq 6$ is false; $6 \geq 6$ and $7 \geq 6$ are true.

Step: 3. Choose one branch

What happens and why: Double 5 to 10; subtract 1 from 6 to 5; subtract 1 from 7 to 6.

Step: 4. Report

What happens and why: The three independent runs report 10,5,6.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

The condition is evaluated after adding 2. Equality belongs to the at-least branch, so starting at 4 checks the boundary. Once a branch finishes, the program reports; it does not reconsider the condition just because doubling 5 produced 10.

What this example does not establish

A large reported value need not mean the true branch ran; branch choice depends on the earlier value.

7. Find and repair a mistake

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case · Evidence · What it shows

Case: Rain = yes

Evidence: then choose raincoat

What it shows: clear binary branch

Case: Temperature < 15 C

Evidence: then choose warm layer

What it shows: numeric threshold

Case: Exactly 15 C

Evidence: needs boundary rule

What it shows: edge case

Case: Rain and cold

Evidence: two conditions overlap

What it shows: order or combined rule needed

Repair two tempting claims

CLAIM A If one example works, the branch is complete.

Repair:

CLAIM B Changing branch order cannot change the result.

Repair:

STRONG ANSWER FRAME I claim _____ because the evidence shows _____. The process is _____. One limit is _____.

A closer explanation

Our examples so far use known whole-number scores. If score is unreadable when the test is reached, we cannot establish whether it is at least 6. Unknown does not mean false, and a blank is not automatically zero. For this paper exercise, pause the procedure and request the missing value before selecting a branch.

A complete procedure must specify what to do when required input is unavailable. Another system might define a separate fallback, but it must be written explicitly. Testing below, at and above the boundary checks comparison behaviour; it does not replace a missing-input policy or prove every possible implementation correct.

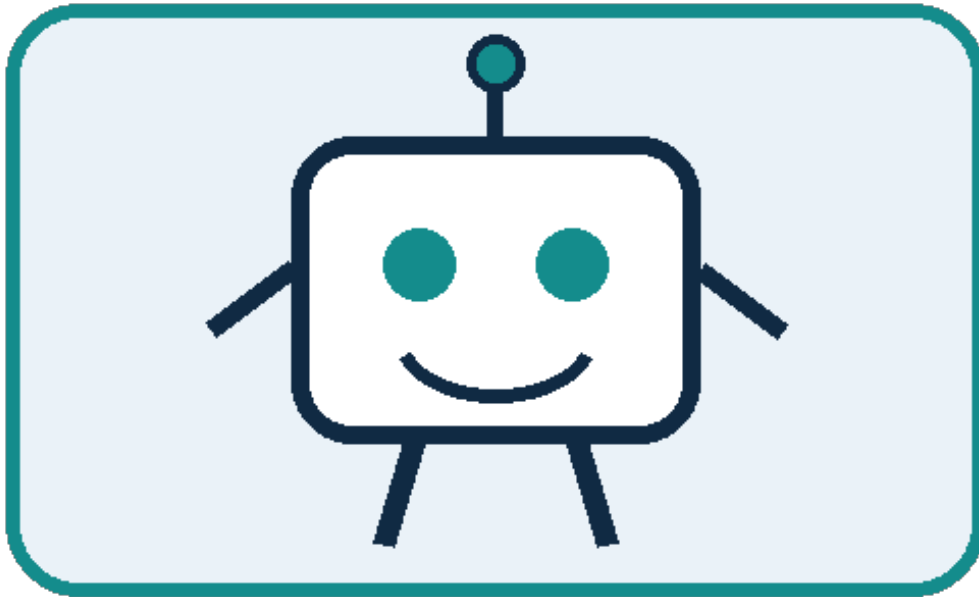
A tester checks only scores 2 and 10 at the condition and says the at-least-6 comparison is correct. Which additional value is most useful for detecting an accidental above-6 comparison, and why

8. Practise together

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: Conditions make an algorithm branch when information changes.

Use a fresh example to explain condition.

Trace a new input → process → output example.

Find and repair the error in: Changing branch order cannot change the result.

Invent one edge case and name the safe response.

PARTNER CHECK Point to the evidence. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

Suppose a copied procedure says above 6 instead of at least 6. A score of 5 at the condition takes the false branch under both versions. A score of 7 takes the true branch under both. But exactly 6 separates them: the intended version subtracts 3 and gives 3; the copied version adds 4 and gives 10.

Choose an initial score that reaches exactly 6 after the first add-2 instruction. Trace both the intended at-least version and the copied above version. Explain why this test reveals an error that the neighbouring values can miss

Record the initial value, the value at the condition, the true or false result, and the chosen update. A final number alone may not reveal whether the test or the arithmetic was wrong. Keep the expected behaviour tied to the stated policy instead of changing it to match a result.

Practise with less help: Read four boundary phrases

Use workbook W2 for the connected example “Conditions and boundary cases”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Conditions and boundary cases. Proposed explanation: “The condition reads the starting value”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Which input detects the difference between the two written conditions? Explain its entire run.

My explanation using a pictured step and the given facts

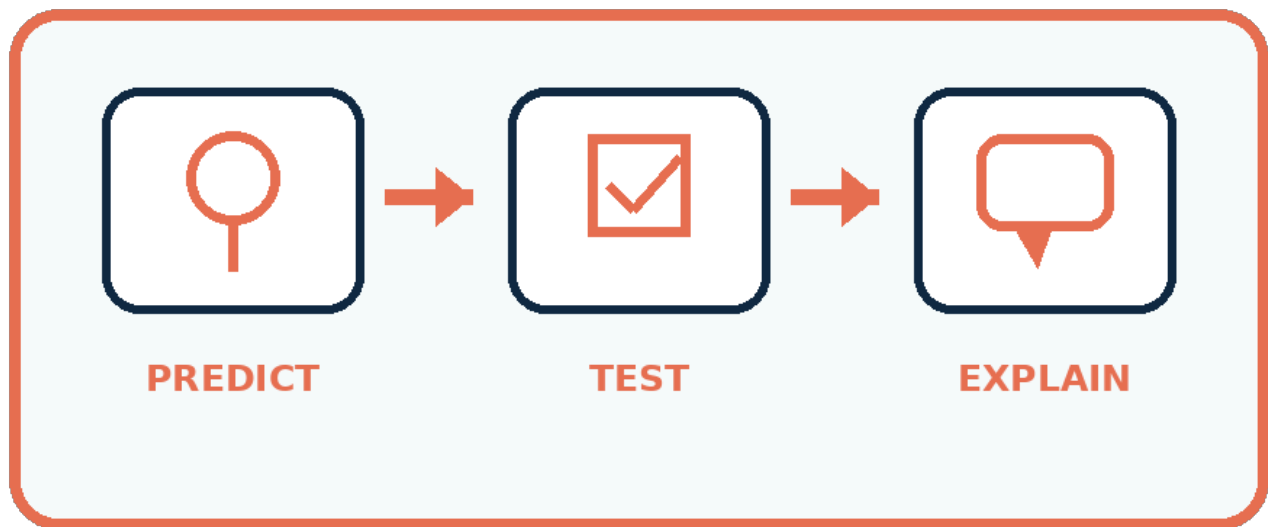
This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EXPLORER LAB

Weather Wardrobe Branches

Predict first, test honestly, and explain what the evidence means.



State your prediction before the result is known.

Complete the task: Build a weather-clothing rule and test four cases.

Keep a normal case and an edge case clearly separated.

Record what happened—even when it does not match your prediction.

Explain the result, one limitation, and the next useful test.

Case · Prediction · Observed result · What it means

Case: Normal

Prediction: _____

Observed result: _____

What it means: _____

Case: Edge

Prediction: _____

Observed result: _____

What it means: _____

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

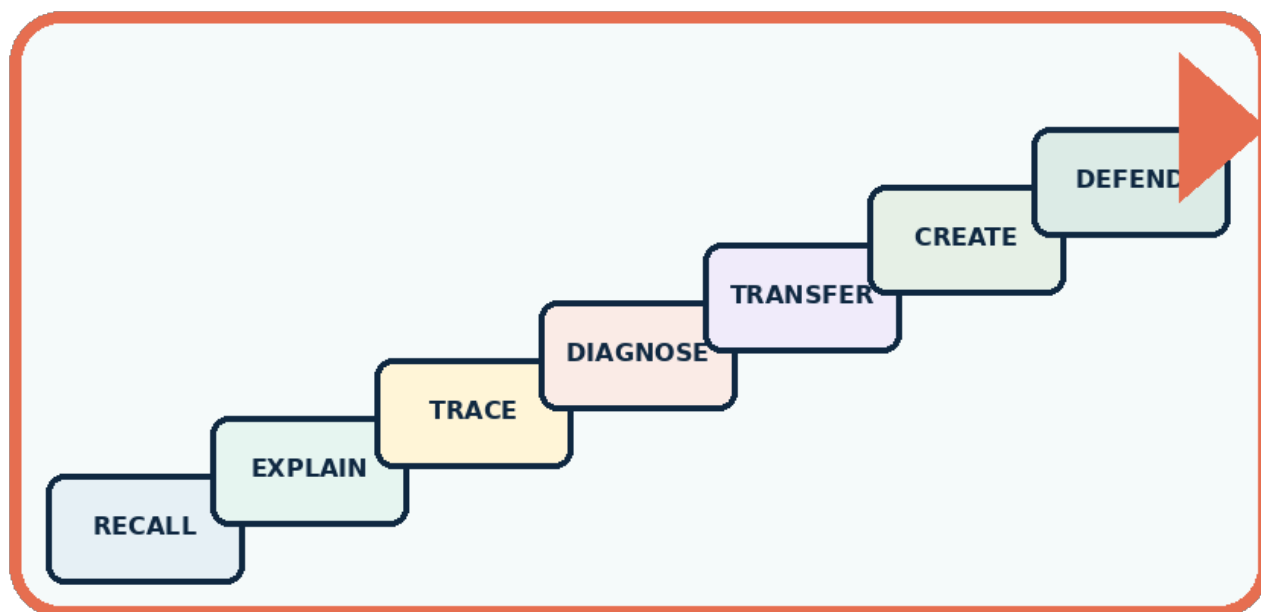
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RECALL Define condition.

EXPLAIN Conditions make an algorithm branch when information changes.

TRACE Show the input, process, and output.

DIAGNOSE Repair: If one example works, the branch is complete.

TRANSFER Apply the idea to a new home or classroom case.

CREATE Design one revealing test.

DEFEND Give a claim, evidence, mechanism, limit, and safe action.

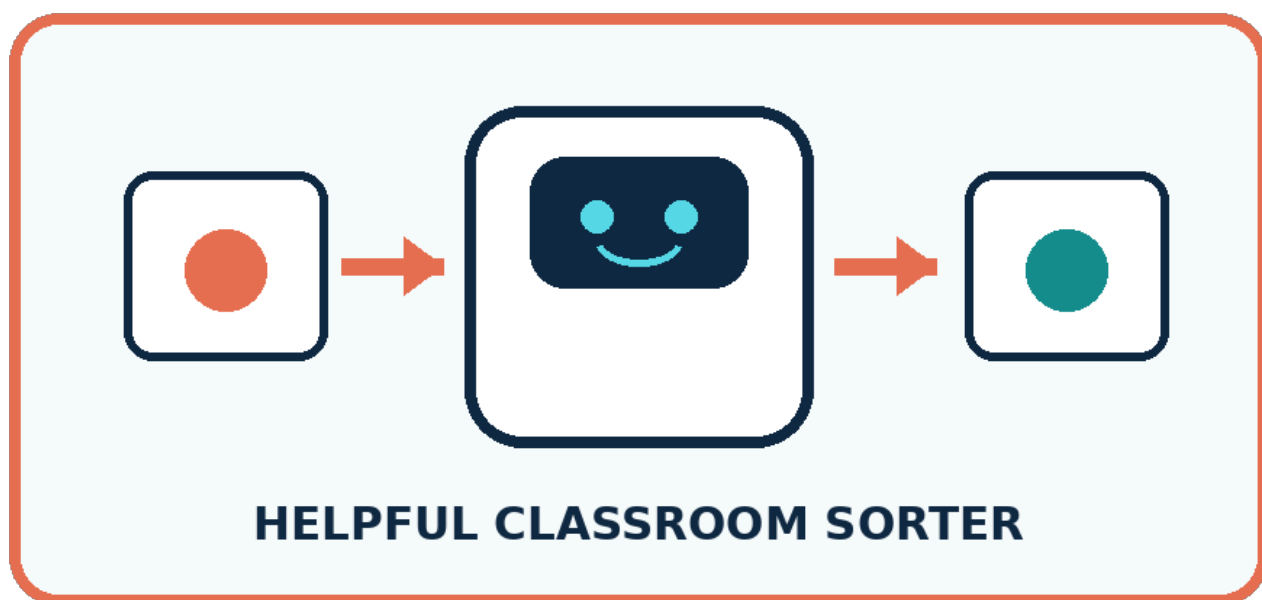
RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

11. Connect your project

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one evidence-based decision at a time.



THIS WEEK'S CHECKPOINT Add one transparent if/then branch and one else path to the Classroom Sorter.

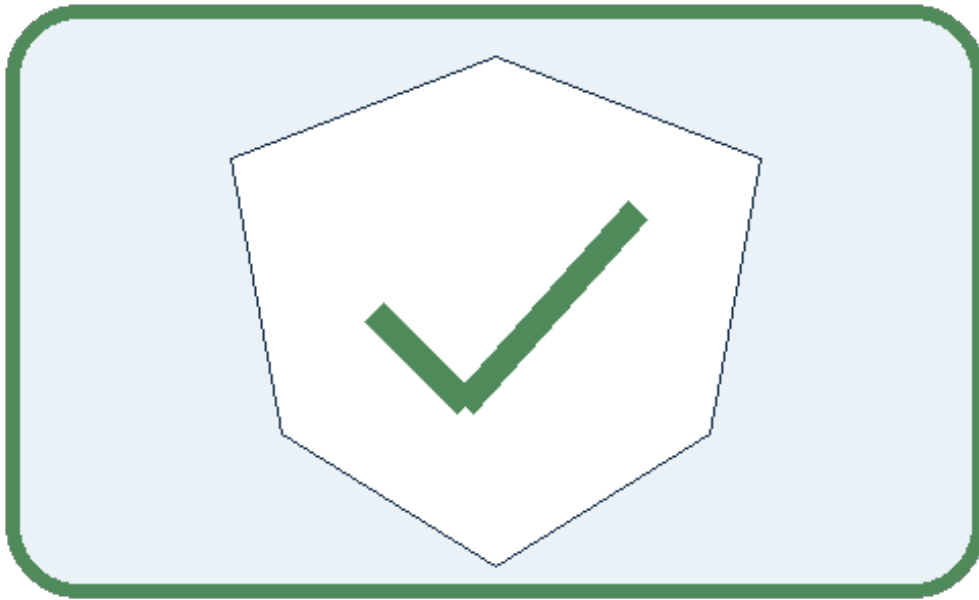
What real classroom need does this serve?

What information is necessary—and what should stay private?

What happens inside the process?

What normal and edge cases will you test?

When should it say not sure or ask a person?



RESPONSIBLE DESIGN Never hide an unhandled case between two thresholds.

My project decision:

Evidence for your project

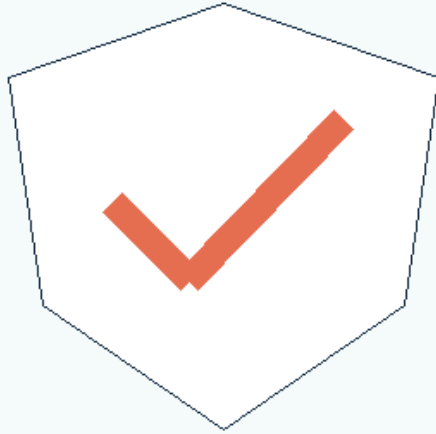
Include equality examples when reviewing sorter conditions, and trace from original input through output.

12. Show what you know

MASTERY

Fresh-Case Independent Proof

Complete this after teaching and guided practice have ended.



INDEPENDENT PROOF

REQUIRED PROOF Build a clothing rule and trace sunny, rainy, cold, and exactly-at-threshold cases.

- ☐ Name or draw the fresh case.
- ☐ Trace input, process, and output.
- ☐ State the conclusion and strongest evidence.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Independent evidence:

SELF-CHECK Each test follows the stated condition order. The exactly-at-threshold case reveals whether the rule uses less than or less-than-or-equal. Rain-plus-cold needs combined outputs or a clear priority.

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

NEXT BRIDGE Carry today's evidence habit into: Repeating Efficiently.

13. Study a complete case

Week 10 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

If a card has a star, send it to A; otherwise send it to B.

The question

Sort a star, a circle, and a card with both shapes.

Worked explanation

Star and mixed cards go to A because both have a star. The circle goes to B. The test is has a star, not has only a star.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Does this card have a star?

Evidence or stage · Value or result

Evidence or stage: Star only

Value or result: Yes → tray A

Evidence or stage: Circle only

Value or result: No → tray B

Evidence or stage: Star and circle

Value or result: Yes → tray A

If a card has a star, send it to A; otherwise send it to B.

Sort a star, a circle, and a card with both shapes.

Star and mixed cards go to A because both have a star. The circle goes to B. The test is has a star, not has only a star.

14. Try a new case independently

Independent case: Transfer the method to a new display

A paper brightness display adds 3 to its starting number. If the current brightness is at most 7, subtract 2; otherwise add 1. Report and stop. Trace separate runs from starting values 3, 4 and 5. Identify the run that tests equality. Then show how its result changes if at most 7 is accidentally written as below 7.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 10 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Use this exact rule: if a card has more than three dots, place it in tray A; otherwise use tray B. Trace cards with 2, 3 and 4 dots.

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Add one transparent if/then branch and one else path to the Classroom Sorter.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

A club admits a visitor if they have a ticket AND an accompanying adult. Test all four yes/no combinations. A student changes AND to OR. Which outcomes change?

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.